

MTU_ValueService **Technical Documentation**

Diesel Engine
V 4000 Px1
Application Group 3A

Maintenance Schedule
M050778/01E



12V4000P61
12V4000P81
16V4000P61
16V4000P81

Table 1: applicable for ...

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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1 Maintenance Schedule

1.1 Preface

MTU maintenance concept

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high level of equipment availability.

The maintenance schedule is based on the load profile / load factor specified below. The time intervals at which the maintenance work is to be carried out and the relevant checks and tasks involved are average values based on operational experience and are therefore to be regarded as guidelines only. Special operating conditions and technical requirements may require additional maintenance work and/or modification of the maintenance intervals. In order to be authorized to carry out the individual maintenance jobs, maintenance personnel must have achieved a level of training and qualification appropriate to the complexity of the task in hand. The various Qualification Levels QL1 to QL4 reflect the levels of training offered in MTU courses and the contents of the tool kits required:

QL1: Operational monitoring and maintenance which can be carried out during a break in operation without disassembling the engine.

QL2: Component exchange (corrective only).

QL3: Maintenance work which requires partial disassembly of the engine.

QL4: Maintenance work which requires complete disassembly of the engine.

The maintenance schedule matrix normally finishes with extended component maintenance. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the list of jobs provide a reference to the scope of parts required.

Notes on maintenance

Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications A001061 and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Amongst other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Fuel prefilter:
The maintenance interval depends on how dirty the fuel is. The paper inserts in fuel prefilters must be changed every two years at the latest (Task 9998).
- Battery:
Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

The relevant manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

This Maintenance Schedule may include components which are not installed on your engine; these may be disregarded.

Out-of-service periods

If the engine is to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Fluids and Lubricants Specifications, MTU Publication No. A001061.

Application Group

3A Continuous operation, unrestricted

Load profile

Load factor	85%	75%	50%	30%
Corresponding operating time	20%	40%	30%	10%

1.2 Maintenance schedule matrix

0-10.500 Operating hours

Item	Limit	Operating hours [h]																	
		Daily	500	1.000	1.500	2.000	2.500	3.000	3.500	4.000	4.500	5.000	5.500	6.000	6.500	7.000	7.500	8.000	8.500
Lubrication points	1 a																		
Emergency air shut-off flaps	1 a																		
Engine oil filter	2 a																		
Engine operation	-	X																	
Fuel filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Centrifugal oil filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Oil indicator filter	6 m			X		X		X		X		X		X		X		X	
Engine mounts	1 a			X		X		X		X		X		X		X		X	
Valve gear	2 a			X		X		X		X		X		X		X		X	
Battery charging generator	1 a					X				X				X				X	
Crankcase breathers	2 a							X						X				X	
Air filters	3 a							X						X				X	
Fuel injectors	9 a													X					
Combustion chambers	9 a													X					
Starter	9 a																		X
Component maintenance	9 a																		X
Fuel delivery pump	9 a																		X
Cylinder heads	9 a																		X
Turbochargers	18 a																		X
Extended component maintenance	18 a																		

w = weeks
m = months
a = years

11.000-21.500 Operating hours

Item	Limit	Operating hours [h]																	
		11.000	11.500	12.000	12.500	13.000	13.500	14.000	14.500	15.000	15.500	16.000	16.500	17.000	17.500	18.000	18.500	19.000	19.500
Lubrication points	1 a																		
Emergency air shut-off flaps	1 a																		
Engine oil filter	2 a																		
Engine operation	-																		
Fuel filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Centrifugal oil filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Oil indicator filter	6 m	X		X		X		X		X		X		X		X		X	
Engine mounts	1 a	X		X		X		X		X		X		X		X		X	
Valve gear	2 a	X		X		X		X		X		X		X		X		X	
Battery charging generator	1 a			X				X				X				X			
Crankcase breathers	2 a			X						X					X				X
Air filters	3 a			X						X					X				X
Fuel injectors	9 a			X											X				
Combustion chambers	9 a			X											X				
Starter	9 a																		X
Component maintenance	9 a																		X
Fuel delivery pump	9 a																		X
Cylinder heads	9 a																		X
Turbochargers	18 a																		X

w = weeks
m = months
a = years

Item	Limit	Operating hours [h]															
		11.000	11.500	12.000	12.500	13.000	13.500	14.000	14.500	15.000	15.500	16.000	16.500	17.000	17.500	18.000	18.500
Extended component maintenance	18 a																
w = weeks m = months a = years																	

22.000-30.000 Operating hours

Item	Limit	Operating hours [h]															
		22.000	22.500	23.000	23.500	24.000	24.500	25.000	25.500	26.000	26.500	27.000	27.500	28.000	28.500	29.000	29.500
Lubrication points	1 a																
Emergency air shut-off flaps	1 a																
Engine oil filter	2 a																
Engine operation	-																
Fuel filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Centrifugal oil filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Oil indicator filter	6 m	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Engine mounts	1 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Valve gear	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Battery charging generator	1 a	X			X			X				X			X		
Crankcase breathers	2 a				X						X				X		
Air filters	3 a				X						X				X		
Fuel injectors	9 a				X										X		
Combustion chambers	9 a				X										X		
Starter	9 a														X		
Component maintenance	9 a														X		
Fuel delivery pump	9 a														X		
Cylinder heads	9 a														X		
Turbochargers	18 a														X		
Extended component maintenance	18 a														X		
w = weeks m = months a = years																	

1.3 Maintenance tasks

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Task
QL1	-	1 a	Lubrication points	Apply lubricant at lubrication points	W1053
QL1	-	1 a	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical)	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years)	W1008
QL1	Daily	-	Engine operation	Check engine oil level Carry out visual inspection of engine for general condition and leaks Inspect intercooler drain system (if fitted) Inspect service indicator of air filter Check the inspection bores of the HP fuel pump Check relief bores of coolant pump(s) Check for abnormal running noises, exhaust gas color and vibration Drain off water and contamination from fuel prefilter (if fitted) Check service indicator of fuel prefilter (if fitted) Inspect battery-charging generator for contamination and clean if necessary	W0500 W0501 W0502 W0503 W0504 W0505 W0506 W0507 W0508 W0525
QL1	500	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert	W1001
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed (if fitted)	W1009
QL1	1000	6 m	Oil indicator filter	Check and clean oil indicator filter	W1047
QL1	1000	1 a	Engine mounts	Carry out visual inspection of engine mounts for general condition	W1463
QL1	1000	2 a	Valve gear	Check valve clearance	W1002
QL1	2000	1 a	Battery charging generator	Remove battery-charging generator and clean thoroughly with compressed air	W1209
QL1	3000	2 a	Crankcase breathers	Crankcase breathers: Fit new filters or filter inserts	W1046
QL1	3000	3 a	Air filters	Fit new air filters	W1005
QL1	6000	9 a	Fuel injectors	Fit new fuel injectors	W1006
QL1	6000	9 a	Combustion chambers	Inspect cylinder chambers using endoscope	W1011
QL3	10000	9 a	Starter	Replace copper beryllium pinion (if fitted), after 300 starts, at the latest	W1684
QL3	10000	9 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces Clean air ducting Clean intercooler and inspect for leakage Fit new high-pressure fuel sensor Check engine coolant thermostat and fit new thermal actuator Check charge-air coolant thermostat and fit new thermal actuator Overhaul engine coolant preheating system (if fitted) Overhaul starter Inspect vibration damper Overhaul charge-air coolant pump Fit new seals/sealing materials for all disassembled components Overhaul engine coolant pump	W2000 W2001 W2002 W2003 W2004 W2006 W2007 W2008 W2010 W2011 W2070 W2062 W2110
QL3	10000	9 a	Fuel delivery pump	Fit new fuel delivery pump	W1051
QL3	10000	9 a	Cylinder heads	Overhaul cylinder heads	W1134
QL3	10000	18 a	Turbochargers	Overhaul turbochargers	W1038
QL4	30000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required Replace all elastomeric parts and seals with new ones Fit new piston rings Fit new conrod bearings Fit new crankshaft bearings	W3000 W3001 W3002 W3003 W3004

w = weeks
m = months
a = years

Qualifi- cation level	Interval	Limit	Item	Maintenance tasks	Task
	[h]				
				Fit new cylinder liners	W3005
				Fit new high-pressure fuel pump	W3007
				Fit new actuator cylinders for exhaust flaps	W3012
				Fit new exhaust flap bearings	W3013
				Fit new pressure relief valve in high-pressure fuel system	W3021
w = weeks m = months a = years					